Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Two copper spheres A and B of the same surface finish are taken. Sphere B weight half as A. Both are heated to the same temperature and let in a room to cool by radiation only. What is the ratio of initial rate of loss of heat of A and B.

(A) 0.166

(B) 15.9

(C) 0.159

(D) 0.79

Q.2

Three samples of the same gas A, B and $C(Y = 3/2)$ have initial equal volumes. Now the volume of each sample is doubled. The process is adiabatic for A, isobaric for B and isothermal for C. If the final pressures are equal for all three samples, the ratio of their initial pressures are

(A) $2\sqrt{2} : 2 : 1$

(B) $2\sqrt{2} : 1 : 2$

(C) $\sqrt{2} : 1 : 2$

(D) $2 : 1 : \sqrt{2}$

Q.3

A metal ball cools from $62^{\circ}\mathrm{C}$ to $50^{\circ}\mathrm{C}$ in $10\ \mathrm{min}$ and to $42^{\circ}\mathrm{C}$ the next 10 minutes. What will be its temperature at the end of next ten minutes ?

(A) $3.67^{\circ}\mathrm{C}$

(B) $36.7^{\circ}\mathrm{C}$

(C) $.376^{\circ}\mathrm{C}$

(D) $367^{\circ}\mathrm{C}$

Q.4

0.271 moles of a monoatomic ideal gas at an initial temperature of $271^{\circ}\mathrm{C}$ is expanded to another state by two alternative processes :
Process A : Constant internal energy process, ratio of volumes $= 2.71$
Alternative process B : Initial and final internal energies are unequal.

@IITJEE_Advanced

Work done in one process is twice that in the other. Also heat supplied in one process is twice that in the other. Determine the final temperature in process B, in $^{\circ}\mathrm{C}$, rounded to nearest integer.

(A) 815

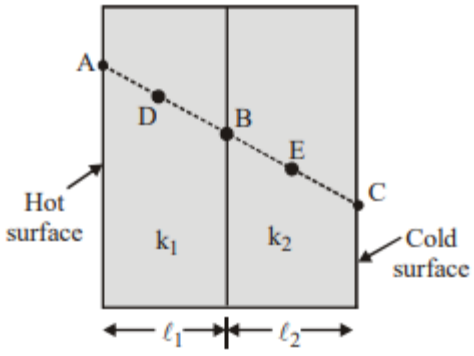
(B) 915

(C) 715

(D) 615

Q.5

Two layers of insulators are kept as shown. Hot surface is maintained at $1000^{\circ}\mathrm{C}$. ABC is an imaginary straight line, D and E are mid points of AB and BC respectively; $\ell_1 < \ell_2$; thermal conductivities are k_1 and k_2 as shown. At steady state temperatures at D and E are $875^{\circ}\mathrm{C}$ and $600^{\circ}\mathrm{C}$. Then $-$



(A) $\mathrm{k}_1 > \mathrm{k}_2$

(B) $\mathrm{k}_1 = \mathrm{k}_2$

(C) $\mathrm{k}_1 = 1.3\mathrm{k}_2$

(D) $\frac{\mathrm{k}_1}{\mathrm{k}_2} < 1.2$

Q.6

A vertical thermally insulated cylinder of volume V contain n moles of an ideal monoatomic gas under a weightless piston. A load of weight W is placed on the piston as a result of which the piston is displaced. If the initial temp of the gas is $300\ \mathrm{K}$, area of piston is A and atmospheric pressure P_0. (take $\mathrm{W} = \mathrm{P}_0\ \mathrm{A}$). Determine the value of final temperature of the gas.

(A) $375\ \mathrm{K}$

(B) $425\ \mathrm{K}$

(C) $475\ \mathrm{K}$

(D) $515\ \mathrm{K}$

Q.7

An ice box made of $1.5\ \mathrm{cm}$ thick Styrofoam has dimensions $60\ \mathrm{cm} \times 60\ \mathrm{cm} \times 30\ \mathrm{cm}$. It contains ice at $0^{\circ}\mathrm{C}$ and is kept in a room $40^{\circ}\mathrm{C}$. Find the rate at which the ice is melting. Latent heat of fusion of ice $= 3.36 \times 10^5\ \mathrm{J/kg}$. and thermal conductivity of styrofoam $= 0.04\ \mathrm{W/m} - {}^{\circ}\mathrm{C}$.

@IITJEE_Advanced

(A) $0.46\ \mathrm{g/s}$

(B) $2.46\ \mathrm{g/s}$

(C) $1.46\ \mathrm{g/s}$

(D) $3.46\ \mathrm{g/s}$

Q.8

A scale, made of a metal of coefficient of linear expansion α, graduated at $0^{\circ}\mathrm{C}$, when used at $\theta^{\circ}\mathrm{C}$, reads a length of ℓ_1 as ℓ_2. Then at same $\theta^{\circ}\mathrm{C}$, it will read a length of ℓ_2 as $-$

(A) $\frac{\ell_1^2}{\ell_2}$

(B) $\frac{\ell_2^2}{\ell_1}$

(C) $\ell_1(1+2\alpha\theta)$

(D) $\ell_2(1+2\alpha\theta)$

Q.9

An ideal gas of 1 mole undergoes a process such that $P \propto \frac{1}{T}$. The molar heat capacity of this process is $33.24\ \mathrm{J/molK}$. Then which of the following statement/s is/are correct?

(A) The work done by the gas is $2\mathrm{R}\Delta\mathrm{T}$.

(B) Degree of freedom of the gas is 4 .

(C) Degree of freedom of the gas is 3 .

(D) $\gamma = \left(\frac{\mathrm{C_P}}{\mathrm{C_V}}\right)$ for the gas is 1.5.

Q.10

A gaseous mixture consists of $8\ \mathrm{g}$ of oxygen and $20\ \mathrm{g}$ of argon. The gases are assumed to be ideal. Then (give molecular weight of argon is 40 and $R = 8.3\ \mathrm{J/mol-K}$)

(A) γ of the gas mixture is 1.55

(B) $\mathrm{C_v}$ of the gas mixture is $15.2\ \mathrm{J\ mol^{-1}\ K^{-1}}$

(C) $\mathrm{C_p}$ of the gas mixture is $23.56\ \mathrm{J\ mol^{-1}\ K^{-1}}$

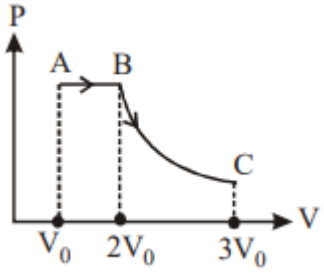
(D) All the above are incorrect

Q.11

One mole of a gas is subjected to two process AB and BC, one after the other as shown in the figure. BC is represented by $\mathrm{PV^n}$ = constant.
We can conclude that (where T = temperature,

@IITJEE_Advanced

W = work done by gas, V = volume and U = internal energy).



(A) $T_A = T_B = T_C$

(B) $V_A < V_B, P_B < P_C$

(C) $W_{AB} > W_{BC}$

(D) $T_A < T_B$

Q.12

Passage:
A $2000cc$ sample of O_2 is confined to a cylinder. Initially, pressure of the gas is $10^5\ N/m^2$ and its temperature $27^\circ C$. The gas is subjected to a cyclic process. In the first step, its pressure is made twice at constant volume. In the second step, it expands to its initial pressure adiabatically and in the final step, the gas undergoes isobaric compression and finally attains its initial volume. Consider that the gas is ideal.
Question: Temperature of the at the end of the first step, i.e. just at the start of adiabatic expansion will be -

(A) $177^\circ C$

(B) $327^\circ C$

(C) $127^\circ C$

(D) $148^\circ C$

Q.13

Passage:
A $2000cc$ sample of O_2 is confined to a cylinder. Initially, pressure of the gas is $10^5\ N/m^2$ and its temperature $27^\circ C$. The gas is subjected to a cyclic process. In the first step, its pressure is made twice at constant volume. In the second step, it expands to its initial pressure adiabatically and in the final step, the gas undergoes isobaric compression and finally attains its initial volume. Consider that the gas is ideal.
Question: Volume of the gas at the end of the second step (adiabatic expansion) will be nearly -

(A) $4124cc$

(B) $1876cc$

(C) $2584cc$

(D) $3272cc$

Q.14

Passage:
A $2000cc$ sample of O_2 is confined to a cylinder. Initially, pressure of the gas is $10^5\ N/m^2$ and its temperature

@IITJEE_Advanced

$27^{\circ} \mathrm{C}$. The gas is subjected to a cyclic process. In the first step, its pressure is made twice at constant volume. In the second step, it expands to its initial pressure adiabatically and in the final step, the gas undergoes isobaric compression and finally attains its initial volume. Consider that the gas is ideal.

Question: During the cycle -

(A) work done by the gas is about $55 \mathrm{~J}$ and heat absorbed by the gas is also about $55 \mathrm{~J}$

(B) work done by the gas is about $42 \mathrm{~J}$ and heat rejected by the gas is $60 \mathrm{~J}$

(C) work done on the gas is $60 \mathrm{~J}$ and heat rejected by the gas is also $60 \mathrm{~J}$

(D) work done on the gas is $60 \mathrm{~J}$ and heat absorbed by the gas is also about $60 \mathrm{~J}$

Q.15

Passage:

Consider air to be a diatomic ideal gas with average mole masses $29 \mathrm{~g} / \mathrm{mole}$. A mass $1.45 \mathrm{~kg}$ of air is contained in a cylinder with a piston at $27^{\circ} \mathrm{C}$ and pressure $1.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. Energy is given to the system as heat and the system is allowed to expand till the final pressure $3.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. As the gas expands, pressure and volume follow the relation, $\mathrm{V}=\mathrm{KP}^{2}$ where K is a constant. $(\mathrm{R}=8.3 \mathrm{~J} /(\text { mole-K }))$

Question: Final volume of the system will be nearly -

(A) $6.2 \mathrm{~m}^{3}$

(B) $2.8 \mathrm{~m}^{3}$

(C) $4.5 \mathrm{~m}^{3}$

(D) $1.4 \mathrm{~m}^{3}$

Q.16

Passage:

Consider air to be a diatomic ideal gas with average mole masses $29 \mathrm{~g} / \mathrm{mole}$. A mass $1.45 \mathrm{~kg}$ of air is contained in a cylinder with a piston at $27^{\circ} \mathrm{C}$ and pressure $1.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. Energy is given to the system as heat and the system is allowed to expand till the final pressure $3.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. As the gas expands, pressure and volume follow the relation, $\mathrm{V}=\mathrm{KP}^{2}$ where K is a constant. $(\mathrm{R}=8.3 \mathrm{~J} /(\text { mole-K }))$

Question: Final temperature of the system will be nearly -

(A) $2125 \mathrm{~K}$

(B) $3800 \mathrm{~K}$

(C) $3200 \mathrm{~K}$

(D) $4250 \mathrm{~K}$

Q.17

Passage:

Consider air to be a diatomic ideal gas with average mole masses $29 \mathrm{~g} / \mathrm{mole}$. A mass $1.45 \mathrm{~kg}$ of air is contained in a cylinder with a piston at $27^{\circ} \mathrm{C}$ and pressure $1.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. Energy is given to the system as heat and the system is allowed to expand till the final pressure $3.5 \times 10^{5} \mathrm{~N} / \mathrm{m}^{2}$. As the gas expands, pressure and volume

@IITJEE_Advanced

follow the relation, $V = KP^2$ where K is a constant. $(R = 8.3\ J/(\text{mole-K}))$

Question: Increase of internal energy of the system is nearly -

(A) 3.6×10^6 J

(B) 4.2×10^4 J

(C) 86×10^3 J

(D) 6.3×10^4 J

Q.18

Match the column -

Column I	Column II
(A) Heat current is directly proportional to cross section area	(p) Conduction
(B) For small temperature difference heat current is directly proportional to temperature difference	(q) Convection
(C) Transfer of heat due to difference in density	(r) Radiation
(D) Fastest heat transfer	(s) Kirchhoff's law

(A) A-(pqr) B-(pqr) C-(r) D-(q)

(B) A-(pqr) B-(pq) C-(r) D-(q)

(C) A-(pqr) B-(pqr) C-(q) D-(r)

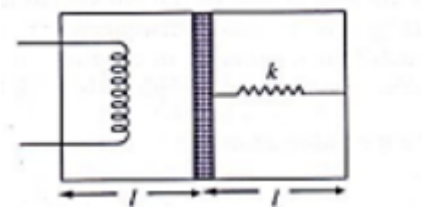
(D) A-(pqr) B-(pq) C-(s) D-(p)

Q.19

The relation between internal energy U, pressure P and volume V of an ideal gas in a adiabatic process is $U = 2 + 3PV$. The value of ratio of the molar specific heats $\left(\frac{C_P}{C_V}\right)$ is $x/3$. Find the value of x.

Q.20

An adiabatic cylinder of length $1\ m$ and area of cross section $10^{-2}\ m^2$ is closed at both ends. A freely moving non conducting thin piston divides the cylinder into two equal parts. Each part contains $28gm$ of N_2. The natural length of the spring connected to the piston and the right wall of the cylinder is $l = 50\ cm$ and $k = \sqrt{2} \times 10^3 \frac{N}{m}$. Initially $\frac{1}{3}$ nd molecules of the nitrogen in the right part are dissociated into atoms. Initial pressure in each part is $P_0 = \sqrt{2} \times 10^5\ N/m^2$. The gas on left side is heated slowly by the heater so as to compress the spring by $\frac{3l}{4}$. The work done by the gas in right part is $-707\ nJ$. Find the value of n.



Answers & Solutions

Q.1 Answer:

@IITJEE_Advanced

0.79

Solution:

Rate of loss of heat is given by

$$\frac{\Delta Q}{\Delta t} = e\sigma AT^4$$

$$\text{also } \Delta Q = mc\Delta T$$

$$\therefore \frac{mc\Delta T}{\Delta t} = e\sigma AT^4$$

$$\text{or } \frac{\Delta T}{\Delta t} = \frac{e\sigma AT^4}{mc}$$

which gives the rate of change of temperature or rate of cooling

For a sphere, mass $\mathrm{m} = \frac{4}{3}\pi r^3\rho$

$$\Rightarrow r = \left(\frac{3m}{4\pi\rho}\right)^{\frac{1}{3}}, A = 4\pi r^2$$

$$\frac{\Delta T}{\Delta t} = \frac{4\pi e\sigma T^4}{c}\left[\frac{1}{m}\left(\frac{3m}{4\pi\rho}\right)^{\frac{2}{3}}\right]$$

$$K\left(\frac{1}{m}\right)^{\frac{1}{3}} \ (\mathrm{K} = \text{ constant })$$

All other quantity are same for both sphere.

$$\frac{\left(\frac{\Delta T}{\Delta t}\right)_A}{\left(\frac{\Delta T}{\Delta t}\right)_B} = \left(\frac{m_B}{m_A}\right)^{\frac{1}{3}} = \left(\frac{1}{2}\right)^{1/3}$$

Q.2 Answer:

$2\sqrt{2} : 1 : 2$

Solution:

Let the initial pressure of the three samples be P_A, P_B **and** P_C**, then**

$$P_A(V)^{3/2} = (2V)^{3/2}P$$

$$P_B = P$$

$$\text{and } P_C(V) = P(2V)$$

$$\therefore P_A : P_B : P_C = (2)^{3/2} : 1 : 2 = 2\sqrt{2} : 1 : 2$$

$$\therefore (\mathrm{b})$$

Q.3 Answer:

$36.7^\circ\mathrm{C}$

Solution:

Use $\frac{\theta_1-\theta_2}{t} = K\left(\frac{\theta_1-\theta_2}{2} - \theta_0\right) ...(1)$

To get the following equations

$\frac{62-50}{10} = K\left(\frac{62+50}{2} - \theta_0\right)$ **and** $\frac{50-42}{10} = \left(\frac{50-42}{2} - \theta_0\right) ...(2)$

Divide 2 by 1 and Solve to get θ_0

$\frac{8}{12} = \frac{46-\theta_0}{56-\theta_0} \Rightarrow \theta_0 = 26 ...(3)$

Let after the next 10 **min the temperature falls to then**

@IITJEE_Advanced

$\frac{42-\theta}{10} = \mathrm{K}\left(\frac{42+\theta}{2} - 26\right) \ldots(4)$

From (3) using value of $\theta_0 = 26$ we get $\frac{12}{10} - \mathrm{k}(56-26) \ldots(5)$

Divide (4) by(5) to get $\frac{42-\theta}{12} = \frac{42+\theta-52}{2(56-26)} \Rightarrow \theta = 36.7^{\circ}\mathrm{C}$

Q.4 Answer:

815

Solution:

$Q = \Delta U + W$

Process A : (Constant $U \Rightarrow$ isothermal)

$\mathrm{Q} = \mathrm{W} = \mathrm{nRT_0} \ln \frac{\mathrm{V_1}}{\mathrm{V_2}} = \mathrm{nRT_0} \ln 2.71 = \mathrm{nRT_0} \ln \mathrm{e} = \mathrm{nRT_0}$

$\mathrm{Process\ B} : \mathrm{Q'} = \Delta \mathrm{U} + \mathrm{W'}, \Delta \mathrm{U} \neq 0$

$\mathrm{Case\ (i)}\ \mathrm{Q'} = 2\mathrm{Q}\ \mathrm{and}\ \mathrm{W'} = 2\ \mathrm{W}$

$\mathrm{Not\ possible} \quad \because \Delta \mathrm{U}\ \mathrm{becomes\ zero}$

Case (ii), $Q' = \frac{Q}{2}, W' = \frac{W}{2} \Rightarrow$ not possible

$\therefore \quad \Delta \mathrm{U}$ **becomes zero**

Case (iii), $Q' = \frac{Q}{2}, W' = 2W \Rightarrow \Delta U = \frac{Q}{2} - 2W = \frac{W}{2} - 2W = -\frac{3}{2}W \Rightarrow \frac{nR\Delta T}{\gamma - 1} = \frac{-3}{2} nRT_0$

$\Rightarrow \quad \Delta \mathrm{T} = \frac{-3}{2}(\gamma - 1)\mathrm{T_0} = -\frac{3}{2}\left(\frac{5}{3} - 1\right)\mathrm{T_0} = -\mathrm{T_0} \Rightarrow \mathrm{T} - \mathrm{T_0} = -\mathrm{T_0}; \mathrm{T} = 0$

not possible, ($\because$ Absolute temperature cannot be zero)

Case (iv), $Q' = 2Q = 2W$ **and** $W' = \frac{W}{2}$

$\Rightarrow \quad \Delta \mathrm{U} = \frac{3}{2}\ \mathrm{W} \Rightarrow$ **Repeating the same steps.**

$\Rightarrow \mathrm{T} = 2\ \mathrm{T_0} = 2 \times 544 = 1088\mathrm{k} = 815^{\circ}\mathrm{C}$

Q.5 Answer:

$\frac{\mathrm{k_1}}{\mathrm{k_2}} < 1.2$

Solution:

$\frac{\mathrm{d}\theta}{\mathrm{dt}} = \mathrm{KA}\frac{\mathrm{d}\theta}{\mathrm{d}\ell} =$ **constant** $\Rightarrow \mathrm{k}\frac{\mathrm{d}\theta}{\mathrm{dt}} =$ **constant**

$\theta_{\mathrm{A}} = 1000, \theta_0 = 875$

$\frac{1000+\theta_{\mathrm{B}}}{2} = 875 \Rightarrow \theta_{\mathrm{B}} = 750^{\circ}\mathrm{C}$

$\theta_{\mathrm{E}} = 600, \quad \therefore \mathrm{k_1} \cdot \frac{250}{\ell_1} = \mathrm{k_2}, \frac{150}{\ell_2/2} = \mathrm{k_2}\frac{300}{\ell_2} \Rightarrow \frac{\ell_1}{\ell_2} = \frac{\mathrm{k_1}}{\mathrm{k_2}} \cdot \frac{250}{300}$

$\frac{\ell_1}{\ell_2} < 1 \Rightarrow \frac{\mathrm{k_1}}{\mathrm{k_2}} \cdot \frac{300}{250} \Rightarrow \frac{\mathrm{k_1}}{\mathrm{k_2}} < 1.2$

Q.6 Answer:

375 K

Solution:

Let T_1 and T_2 be the initial and final temperature, then

$P_0 V = nRT_1$

$\left(P_e + \frac{W}{A}\right)(V - Ah) = nRT_2$

Where h **is the displacement of piston.**

Also, work done on gas $= \Delta \mathrm{U}$

$Wh = nC_v\Delta T = \frac{3}{2}nR\left(T_2 - T_1\right)$

@IITJEE_Advanced

From equation (i). (ii) and (iii)

$$\frac{2}{3}Wh = \frac{WV}{A} - Wh - P_0Ab$$

$$\text{or, } Ah = \frac{WV}{P_0A + \frac{5}{3}W}$$

$$T_2 = \left(P_0 + \frac{W}{A}\right)\left(V - \frac{WV}{P_0A + \frac{5}{3}W}\right)$$

$$T_2 = \frac{5}{4}T_1 = \frac{5}{4} \times 300 = 5 \times 75K = 375K.$$

Q.7 Answer:

$0.46\ \text{g/s}$

Solution:

The total surface area of the walls $= 2(60\ \text{cm} \times 60\ \text{cm} + 60\ \text{cm} \times 30\ \text{cm} + 60\ \text{cm} \times 30\ \text{cm}) = 1.44\ \text{m}^2.$

The thickness of the walls $= 1.5\ \text{cm} = 0.015\ \text{m}.$

The rate of heat flow into the box is

$$\frac{\Delta Q}{\Delta T} = \frac{KA(\theta_1 - \theta_2)}{\text{x}} = \frac{(0.04\ \text{W/m}-^\circ\text{C})\left(1.44\ \text{m}^2\right)(40^\circ\text{C})}{0.015\ \text{m}} = 154\ \text{W}$$

The rate at which the ice melts is $\frac{154\ \text{W}}{3.36 \times 10^5\ \text{J/kg}} = 0.46\ \text{g/s}.$

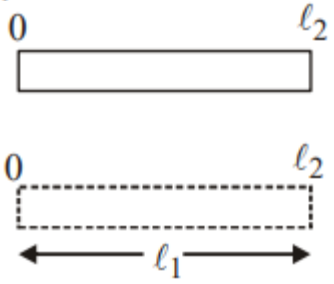
Q.8 Answer:

$\frac{\ell_2^2}{\ell_1}$

Solution:

At $0^\circ\text{C}, \ell_1 = \ell_2(1 + \alpha\theta)$

Similarly, $\ell_2 = \ell_3(1 + \alpha\theta) \Rightarrow \frac{\ell_1}{\ell_2} = \frac{\ell_2}{\ell_3} \Rightarrow \ell_3 = \frac{\ell_2^2}{\ell_1}$



Q.9 Answer:

The work done by the gas is $2\text{R}\Delta\text{T}$.

Solution:

Given $\text{PT} =$ **Constant and** $\text{PV} = \text{nRT}$

$\Rightarrow \text{P}^2\ \text{V} =$ constant

$\Rightarrow \text{PV}^{1/2} = \text{K}$

From first law of thermodynamics

@IITJEE_Advanced

$\Delta Q = \Delta U + W$

$C\Delta T = C_v \Delta T + \left(\frac{P_f V_f - P_1 V_1}{1 - \frac{1}{2}}\right)$

$C = C_v + 2R$

$33.24 = \frac{R}{Y - 1} + 2R$

$\Rightarrow Y = 1.5$

and $Y = 1 + \frac{2}{f}$

$\therefore f = 4$

Q.10 Answer:

γ of the gas mixture is 1.55

Solution:

Let us find γ of the gas mixture which is given by $\frac{n_1+n_2}{\gamma-1} = \frac{n_1}{\gamma_1-1} + \frac{n_2}{\gamma_2-1}$, where n stands for mass in moles.

Since oxygen is diatomic $\gamma_1 = \frac{7}{5}$

For argon $\gamma_2 = \frac{5}{3}$, since argon is monatomic.

Hence, $n_1 = \frac{8}{32} = \frac{1}{4}, n_2 = \frac{20}{40} = \frac{1}{2}$ $\quad \therefore \frac{0.25+0.5}{\gamma-1} = \frac{0.25}{\frac{7}{5}-1} + \frac{0.5}{\frac{5}{3}-1}$

Simplifying $\gamma = 1.55$

$C_v = \frac{R}{\gamma-1} = \frac{8.3}{1.545-1} = 15.2 \text{ J mol}^{-1} \text{ K}^{-1}$

$Cp = \gamma C_v = 1.55 \times 15.2 = 23.56 \text{ J mol}^{-1} \text{ K}^{-1}$

Q.11 Answer:

$W_{AB} > W_{BC}$

Solution:

By $PV = nRT$

$T_B > T_A$

(From A to B, pressure is constant and volume is increasing so temperature also increase)

$W_{AB} = P_0 (2 V_0 - V_0) = P_0 V_0$

$W_{BC} = RT \ln(3/2) = 2P_0 V_0 [\ln 3 - \ln 2]$

$= 2P_0 V_0 (2.303)(0.477 - 0.30) = 0.8142 P_0 V_0 < W_{AB}$

Q.12 Answer:

$327^\circ C$

Solution:

$P \propto T$

$\frac{P_2}{P_1} = \frac{T_2}{T_1};$

$\frac{2P}{P} = \frac{T_2}{300 \text{ K}}; T_2 = 600 \text{ K} = 327^\circ C$

Q.13 Answer:

3272cc

@IITJEE_Advanced

Solution:

$r = 7/5, P_1 = 2P, P_2 = P,$
$\mathrm{PV^r} = \text{ constant}$
$\mathrm{P_1\ V_1r = P_2\ V_2r}$
$\mathrm{2P[2000]^r = PV_2{}^r} \quad [\text{Given: } r = 7/5, P_1 = 2P, P_2 = P]$
$\mathrm{2^r \times 2000 = \frac{1}{2}}$
$\mathrm{v_2 = (2)^{7/5} \times 2000 = 3272\ cc}$

Q.14 Answer:

work done by the gas is about $55\ \mathrm{J}$ and heat absorbed by the gas is also about $55\ \mathrm{J}$

Solution:

Work done by gas during the process is $55\ \mathrm{J}$ and heat absorbed by gas is $55\ \mathrm{J}$.

Q.15 Answer:

$4.5\ \mathrm{m^3}$

Solution:

$\mathrm{PV = nRT}$
Initial volume $= \frac{1.45\times10^3}{29} \times 8.3 \times 300\ \mathrm{m^3}$
$\frac{\mathrm{V_i}}{\mathrm{P_i^2}} = \frac{\mathrm{V_f}}{\mathrm{P_f^2}} \Rightarrow \mathrm{V_f} = \left(\frac{3.5}{1.5}\right)^2\ \mathrm{V_i} = 4.5\ \mathrm{m^3}$
$\mathrm{T_f} = \frac{\mathrm{P_fV_f}}{\mathrm{nR}} = 3800\ \mathrm{K}$
$\Delta\mathrm{U} = \mathrm{nC_V}\Delta\mathrm{T} = \mathrm{n}\frac{5\mathrm{R}}{2}\Delta\mathrm{T} = 3.6 \times 10^6\ \mathrm{J}$

Q.16 Answer:

$3800\ \mathrm{K}$

Solution:

$\mathrm{PV = nRT}$
Initial volume $= \frac{1.45\times10^3}{29} \times 8.3 \times 300\ \mathrm{m^3}$
$\frac{\mathrm{V_i}}{\mathrm{P_i^2}} = \frac{\mathrm{V_f}}{\mathrm{P_f^2}} \Rightarrow \mathrm{V_f} = \left(\frac{3.5}{1.5}\right)^2\ \mathrm{V_i} = 4.5\ \mathrm{m^3}$
$\mathrm{T_f} = \frac{\mathrm{P_fV_f}}{\mathrm{nR}} = 3800\ \mathrm{K}$
$\Delta\mathrm{U} = \mathrm{nC_V}\Delta\mathrm{T} = \mathrm{n}\frac{5\mathrm{R}}{2}\Delta\mathrm{T} = 3.6 \times 10^6\ \mathrm{J}$

Q.17 Answer:

$3.6 \times 10^6\ \mathrm{J}$

Solution:

$\mathrm{PV = nRT}$
Initial volume $= \frac{1.45\times10^3}{29} \times 8.3 \times 300\ \mathrm{m^3}$
$\frac{\mathrm{V_i}}{\mathrm{P_i^2}} = \frac{\mathrm{V_f}}{\mathrm{P_f^2}} \Rightarrow \mathrm{V_f} = \left(\frac{3.5}{1.5}\right)^2\ \mathrm{V_i} = 4.5\ \mathrm{m^3}$
$\mathrm{T_f} = \frac{\mathrm{P_fV_f}}{\mathrm{nR}} = 3800\ \mathrm{K}$
$\Delta\mathrm{U} = \mathrm{nC_V}\Delta\mathrm{T} = \mathrm{n}\frac{5\mathrm{R}}{2}\Delta\mathrm{T} = 3.6 \times 10^6\ \mathrm{J}$

@IITJEE_Advanced

Q.18 Answer:

A-(pqr) B-(pqr) C-(q) D-(r)

Solution:

$(A)-pqr,(B)-pqr,(C)-q,(D)-r$

For conduction, convection & radiation $\frac{\mathrm{dQ}}{\mathrm{dt}} \propto \mathrm{A} \propto \Delta \mathrm{T}$ **(for small temp. difference)**

Free convection is due to difference in density.

Kirchhoff's law : Good emitters are good absorbers.

Q.19 Answer:

—

Solution:

$\mathrm{U}=2+3 \mathrm{nRT}$

$\mathrm{dU}=\mathrm{n} 3 \mathrm{RdT}=\mathrm{nC_V dT} ; \mathrm{C_p}=3 \mathrm{R}+\mathrm{R} ; \frac{\mathrm{C_p}}{\mathrm{C_v}}=\frac{3+1}{3}=\frac{4}{3}$

Q.20 Answer:

—

Solution:

$$
\begin{aligned}
&V_0=Al=5 \times 10^{-3} \mathrm{~m}^3 \\
&P_0=\sqrt{2} \times 10^5 \mathrm{~N} / \mathrm{m}^2 \\
&n_1=\frac{m}{M}=\frac{28}{28}=1 \\
&n_2=2 \times \frac{1}{3}+\frac{2}{3}=\frac{4}{3}\left(n_1 \text{ \& } n_2 \text{ are the number of moles}\right) \\
&C_{n_1}=\frac{5}{2} R, C_h=\frac{7}{2} R \\
&C_{v_2}=\frac{\frac{2}{3} \times \frac{3}{2} R+\frac{2}{3} \times \frac{5}{2} R}{\frac{2}{3}+\frac{2}{3}}=2 R \\
&C_{p_1}=C_{v_2}+R=3 R \\
&\gamma_2=\frac{C_{p_2}}{C_{v_2}}=\frac{3}{2}=1.5
\end{aligned}
$$

$V_1=A \times \frac{7 l}{4}=10^{-2} \times \frac{7}{4} \times 0.5$

$V_1=8.75 \times 10^{-3} \mathrm{~m}^3$

$V_2=2 V_0-V_1=1.25 \times 10^{-3} \mathrm{~m}^3$

$P_2=P_0\left(\frac{V_0}{V_2}\right)^\gamma=P_0\left(\frac{5}{1.25}\right)^{3 / 2}$

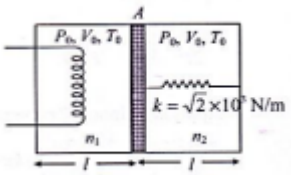
$P_2=8 P_0$

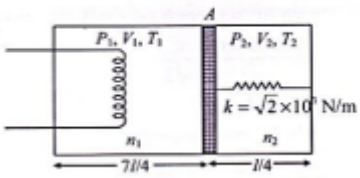
$\Delta W=\frac{P_2 V_2-P_0 V_0}{1-\gamma}$

$\Delta W=-1414 \mathrm{~J}=-707 \times 2 \mathrm{~J}$. So, $n=2$

@IITJEE_Advanced

$\therefore$





@IITJEE_Advanced